JONATHAN LI

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RESEARCH INTERESTS

- I study the interplay among incentives (economics), algorithms (computer science), and learning (statistics), with interests in reinforcement learning, reasoning, search, planning, online learning, and multi-agent systems.
- I develop self-improving large language model (LLM) agents that scale reasoning and decision-making through reinforcement learning (RL), post-training, search, reward modeling, synthetic data, and adaptive inference, with an emphasis on coding agents, multi-agent collaboration, and acquiring capabilities in open-ended environments.

EDUCATION

RENSSELAER POLYTECHNIC INSTITUTE — **Troy, NY**
Ph.D. in Computer Science (GPA: 3.92/4.0) — **Aug 2023 – Present**

The University of Chicago — **Chicago, IL**
Master of Science in Financial Mathematics (GPA: 3.82/4.0) — **Aug 2021 – Mar 2023**

REED COLLEGE — **Portland, OR**
Bachelor of Science in Mathematics and Economics (GPA: 3.93/4.0) — **Aug 2017 – May 2021**

- Completed extensive computer science coursework, including the full major curriculum and nine additional courses.

SELECTED PUBLICATIONS

* denotes equal contribution

1. "Actor-Curator: Co-adaptive Curriculum Learning via Policy-Improvement Bandits for RL Post-Training". Zhengyao Gu*, Jonathan Light*, Raul Astudillo, Ziyu Ye, Langzhou He, Henry Peng Zou, Wei Cheng, Santiago Paternain, Philip S. Yu, Yisong Yue. *COLM 2026*.
2. "DISC: Dynamic Decomposition Improves LLM Inference Scaling". Jonathan Light, Wei Cheng, Benjamin Riviere, Wu Yue, Masafumi Oyamada, Mengdi Wang, Yisong Yue, Santiago Paternain, Haifeng Chen. *NeurIPS 2025*.
3. "Scattered Forest Search: Smarter Code Space Exploration with LLMs". Jonathan Light, Yue Wu, Yiyou Sun, Wenchao Yu, Yanchi Liu, Xujiang Zhao, Ziniu Hu, Wei Cheng. *ICLR 2025*.
4. "Strategist: Self-improvement of LLM Decision Making via Bi-Level Tree Search". Jonathan Light, Henry Cai, Weiqin Chen, Guanzhi Wang, Xiusi Chen, Wei Cheng, Yisong Yue, Ziniu Hu. *ICLR 2025*. Web app. Featured in the State of AI Report alongside o1.
5. "AvalonBench: Evaluating LLMs Playing the Game of Avalon". Jonathan Light*, Henry Cai*, Sheng Shen, Ziniu Hu. *NeurIPS 2023 Workshop on Foundation Models for Decision Making*. Codebase.
6. "A Data-Centric Online Market for Machine Learning: From Discovery to Pricing". Sainyam Galhotra*, Minbiao Han*, Jonathan Light*, Steven Xia*, Raul Castro Fernandez, Haifeng Xu. *ICML 2026*.

RESEARCH EXPERIENCE

MICROSOFT RESEARCH — **Montreal, Canada**
Research Intern, Agentic Coding — **Jun 2026 – Present**

- Developing synthetic-data methods for generating coding tasks and environments for reinforcement learning
- Investigating coding-agent training and credit assignment across multi-step trajectories
- Building distributed, asynchronous RL infrastructure for large-scale agent training and evaluation

ASARI AI — **San Francisco, CA**
Founding Member of Technical Staff, LLM Agents & Training Infrastructure — **May 2025 – Jun 2026**

- Developed GPU-kernel coding agents using world-model predictions to guide iterative search
- Built massively parallel self-improvement infrastructure on Kubernetes, doubling code-translation reliability
- Built observability and evaluation infrastructure for scalable debugging, failure analysis, and reproducibility
- Architected production-scale tool-using LLM agents for reliable, long-horizon C-to-Rust migration

CALTECH, CMS – ADVISOR: YISONG YUE — **Pasadena, CA**
Visiting Researcher, LLM Reasoning and Inference Scaling — **Jan 2025 – Present**

- Developed DISC, an inference-time scaling method that dynamically allocates search to difficult reasoning steps

- Designed an unsupervised method for decomposing LLM reasoning traces into structured subproblems
- Improved base-model accuracy by up to 4× through search, outperforming long-CoT models at matched budgets

RPI COMPUTER SCIENCE – ADVISORS: SANTIAGO PATERNAIN AND ZINIU HU — **Troy, NY**
Ph.D. Researcher, Self-Improving LLM Agents — **Jul 2023 – Present**

- Developed Strategist, enabling human-level LLM-agent performance in Avalon via self-play and self-improvement
- Created AvalonBench, an eval suite for reasoning, communication, and social deduction under partial information
- Designed multi-agent systems with memory, planning, reflection, and search, yielding emergent collaboration

NEC LABORATORIES AMERICA – ADVISOR: WEI CHENG — **Princeton, NJ**
Part-time Researcher, Code Generation — **May 2024 – May 2025**

- Achieved SOTA on HumanEval, LeetCode, and APPS, improving code-generation accuracy by over 5.5%
- Developed test-time scaling and search methods for LLM code generation, surpassing prior methods
- Discovered a novel prompt-perturbation technique that increased LLM exploration and solution diversity

UCHICAGO CS – ADVISORS: HAIFENG XU AND RAUL CASTRO FERNANDEZ — **Chicago, IL**
Market and Mechanism Design Research Project — **Aug 2022 – Jun 2023**

- Developed privacy-aware mechanisms for pricing ML models and data, resulting in an ICML 2026 paper

BOOTH SCHOOL OF BUSINESS – ADVISOR: DACHENG XIU — **Chicago, IL**
Econometrics and Statistics Research Project — **Aug 2021 – Jul 2023**

- Optimized time-series simulation code from Python to C++, reducing runtime by 95%
- Developed Monte Carlo–based online asset-pricing algorithms, doubling benchmark performance

REED COLLEGE – ADVISORS: FELIPE CARRERA AND JONATHAN WELLS — **Portland, OR**
Undergraduate Research and Thesis on Game Theory and Probability — **Aug 2020 – May 2021**

- Studied coalition formation, reinforcement learning, and Monte Carlo methods for an undergraduate thesis

SELECT HONORS, AWARDS, AND SERVICE

Outstanding Paper Award, ICLR 2026 Workshop on Logical Reasoning, 2026
ICML Gold Reviewer, 2026; ICLR Reviewer, 2025; NeurIPS Reviewer, 2025
Member of Phi Beta Kappa, 2021
Reed Commendation for Excellence in Scholarship, 2018, 2019, 2020, 2021
Reed Science Research Fellow, 2020; Reed Financial Services Fellow, 2019

OTHER EXPERIENCES

BOOTH SCHOOL OF BUSINESS (UNIVERSITY OF CHICAGO) — **Chicago, IL**
Teaching Assistant — **Mar 2022 – Jul 2023**

- Designed coursework and delivered lectures for Executive MBA students on real-world AI applications

REED RESEARCH REACTOR — **Portland, OR**
Supervisor and Senior Reactor Operator (NRC Licensed) — **Sep 2017 – May 2021**

- Supervised reactor operations and emergency response, conducting neutron irradiation and gamma spectroscopy projects

SELECTED COURSEWORK

* denotes courses for which I served as a teaching assistant

- **Math/Stats:** Real analysis*, probability theory*, stochastic processes, generalized linear models, modern statistics*
- **CS/ML:** Algorithms*, computer systems, RL*, deep learning, algorithmic game theory, ML theory, complexity theory
- **Economics/Finance:** Econometrics*, option pricing*, market design, market microstructures, portfolio theory

SKILLS AND INTERESTS

Programming languages and tools: Python, C++, R, PyTorch, Kubernetes, Go, SQL, Mathematica
Human languages: English (native speaker), Chinese (native speaker)
Interests: Board games, strategy games, card games, hidden-identity games, tabletop RPGs, game design, fencing, archery, squash, music composition, Ultimate Frisbee, badminton, table tennis